Land Surface Temperature Retrieval From Hyperspectral Thermal Infrared Data Using Improved ResNet and ISSTES Algorithm

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***Abstract*—Land surface temperature (LST) is a crucial variable in the Earth's surface system, playing a key role in understanding the exchanges of material and energy between the surface and the atmosphere. Hyperspectral thermal infrared (TIR) data provide new opportunities for developing methods to retrieve LST from satellite observations. However, the typical physical hyperspectral TIR LST retrieval methods are limited by their reliance on accurate atmospheric correction and specific assumptions, which would introduce complexity and reduce applicability. To address these challenges, this study presents a novel LST retrieval framework that combines a deep residual regression network (DR2N) with the iterative spectrally smooth temperature and emissivity separation (ISSTES) algorithm, refined by Hampel filtering. In this framework, DR2N is trained on simulated data to efficiently retrieve atmospheric parameters, including upwelling radiance, downwelling radiance, and transmissivity, and after that, the refined ISSTES algorithm is applied to simulated data covering various surface types, yielding an overall root mean square error (RMSE) of 1.89 K and a bias of −0.19 K. Subsequently, to further verify the performance of the proposed algorithm, LSTs over four study areas—Spain, North Africa, Hulunbuir, and the Yellow Sea—are retrieved and are compared with infrared atmospheric sounding interferometer (IASI) L2 surface temperature products originating from the European Organization for the Exploitation of Meteorological Satellites (EUMETSAT), showing an RMSE of 0.51 K and a bias of −0.25 K. This confirms the efficacy of the proposed LST retrieval approach.**

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***Index Terms*—Deep residual network, Hampel filter, iterative spectrally smooth temperature and emissivity separation (ISSTES), land surface temperature (LST), thermal infrared (TIR) remote sensing.**

I. Introduction

LAND surface temperature (LST) is a key variable in the Earth's surface system and is crucial for understanding material and energy exchanges between the surface and the atmosphere, as well as climate change [1], [2], [3]. Thermal infrared (TIR) remote sensing provides an effective way to obtain LSTs at global or regional scales. Since the 1960s, various space-borne multispectral sensors have been launched, and researchers have made significant progress in solving the ill-posed radiation transfer equation (RTE) to separate the LST and land surface emissivity (LSE), such as the temperature and emissivity separation (TES) method, the physics-based day/night operational method, and the split-window method [2], [4], [5], resulting in more than 30 types of LST products being publicly available, including those from the Landsat, the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER), the Moderate Resolution Imaging Spectroradiometer (MODIS), the Land Surface Analysis Satellite Application Facility (LSA-SAF), and the Geostationary Operational Environmental Satellite-R Series (GOES-R) [2].

Hyperspectral TIR observations acquired from instruments such as the infrared atmospheric sounding interferometer (IASI) and the cross-track infrared sounder (CrIS) provide extensive insights into atmospheric and surface characteristics by covering thousands of spectral channels. It offers a new direction for developing methodologies to retrieve LST from satellite observations by incorporating new constraints or adopting simplified assumptions [6]. For example, taking advantage of the characteristics that LSE spectra exhibit greater smoothness compared to those containing atmospheric spectral features, Borel [7] proposed the iterative spectrally smooth temperature and emissivity separation (ISSTES) method to simultaneously estimate both LST and LSE, which presents satisfactory performance. Other algorithms, such as the downwelling radiance residual index (DRRI) [8], the correlation-based TES algorithm (CBTES) [9], the stepwise refining TES algorithm (SRTES) [10], the linear spectral emissivity constraint method (LSEC) [11], and the wavelet transform temperature and emissivity separation

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(WTTES) [12], could also be used to accurately retrieve LST or constrain the solution space, thereby regularizing the underdetermined problem. These physical LST retrieval methods described above are usually designed to address the challenges of pathological retrieval by increasing equation constraints or reducing the number of unknowns. However, these methods heavily rely on accurate atmospheric corrections, particularly the estimation of atmospheric downwelling radiance. Those traditional image-based atmosphere correction techniques, such as automatic atmospheric compensation [13] and in-scene atmospheric compensation [14], depend on the stringent assumptions of uniform atmospheric conditions or near-blackbody surfaces [15]. These assumptions would render such methods invalid when applied to medium-to-low spatial resolution hyperspectral data [16]. In addition, radiative transfer models used for atmospheric correction, such as MODTRAN [17] or RTTOV [18], require high-quality atmospheric profile data, which are difficult to obtain [19], [20]. This would restrict the applicability of LST retrieval from hyperspectral thermal data.

Neural networks, especially deep learning, have emerged as a promising approach for LST retrieval due to the ability to fully learn and characterize quantitative relationships between input data, offering a unique advantage in addressing high-complexity, nonlinear pathological computational problems [21]. This capability is expected to exploit the fine spectra of hyperspectral data to explore possible deep connections between variables. The neural network method has been utilized for retrieving atmospheric parameters or LST from hyperspectral data obtained by IASI, CrIS, and other onboard hyperspectral TIR sensors. Existing studies have employed networks such as multilayer perceptron (MLP), radial basis function (RBF), and transformer networks [22], [23], [24], [25], demonstrating their capability to retrieve LST with acceptable accuracies of root mean square errors (RMSEs) lower than 3 K and greater computational efficiency. However, the direct construction of the relationship between observed data and LST using a neural network not only lacks physical interpretability but also heavily depends on the quality of the training data, potentially leading to issues such as insufficient generalization [26]. Therefore, in this study, a novel LST retrieval framework, combining the deep learning mechanism with the interpretability of the physical LST retrieval method, is proposed. In this framework, a deep residual regression network (DR2N), which is a variant of the traditional MLP, is used to initially obtain a rough estimate of the atmospheric parameters (upwelling radiance, downwelling radiance, and transmissivity) without other auxiliary data or real-time radiative transfer calculations, and then, the refined ISSTES algorithm is proposed to achieve accurate and efficient LST estimation from hyperspectral TIR data.

Unlike traditional MLPs, which suffer from vanishing gradients in deep architectures, DR2N integrates residual connections to address this issue. These connections enable DR2N to maintain high accuracy as the network deepens, avoiding the characteristic initial-rise-then-decline pattern of traditional MLPs. The connections also enhance gradient flow during training, leading to faster convergence and a better ability to capture the intricate relationships within the hyperspectral data. By learning residuals between layers rather than the full mapping, DR2N achieves superior accuracy and training efficiency, making it particularly effective for atmospheric correction of hyperspectral TIR data.

The rest of this study is organized as follows. In Section II, the LST retrieval algorithm is presented, Section III introduces the experimental data and research areas, and Section IV provides an exposition of the results and discussion. The conclusions are given in Section V.

II. Methodology

A. Radiation Transfer Model

According to the theory of TIR radiative transfer, neglecting the atmospheric scattering effects, the TIR radiance received by sensors in channel $i$can be written as

$$L_i = L_{s,i}\tau_i + L_{\mathrm{up},i} \tag{1}$$

$$L_{s,i} = \varepsilon_i B_i(T_s) + (1-\varepsilon_i) L_{\mathrm{down},i} \tag{2}$$

where L_i and $L_{s,i}$ are the radiances measured at sensor and at ground (surface-leaving radiance) in channel i, respectively; τ_i is the atmosphere transmissivity in channel i; $L_{\mathrm{up},i}$ and $L_{\mathrm{down},i}$ are upwelling and downwelling atmospheric radiances in channel i, respectively ; ε_i is the LSE in channel i; T_s is LST; and B_i is Planck's function with a temperature of T_s, whose formulation is given as follows:

$$B_i(T) = \frac{C_1}{\lambda_i^5}\left(e^{\frac{C_2}{\lambda_i T}} - 1\right)^{-1} \tag{3}$$

where λ_i is the wavelength in μm, and C_1 and C_2 are constants, respectively ($C_1 = 1.919 \times 10^8 W \cdot \mu m^4 \cdot sr^{-1} \cdot m^{-2}$ and $C_2 = 1.439 \times 10^4 \mu m \cdot K$). As mentioned, the inverse problem governing LST retrieval from TIR radiance is fundamentally ill-posed, with the $N+1$ unknown parameters exceeding the N available radiance measurements. Therefore, the emphasis for retrieving LST from hyperspectral TIR data lies in regularizing this ill-posed problem through either parameter space reduction or supplementary constraint equations.

B. LST Retrieval Algorithm

In this study, the hyperspectral TIR LST retrieval algorithm is implemented in two sequential stages: atmospheric correction using atmospheric upwelling radiance and transmissivity, and LST retrieval using surface-leaving radiance, as outlined in Fig. 1. In this procedure, the atmospheric correction is carried out using DR2N, which is repeatedly trained and validated on simulated datasets generated by 4A/OP, the operational release of the Automatized Atmospheric Absorption Atlas (4A) model. The optimal model is selected based on its retrieval accuracy, and its performance on the retrieval of atmospheric parameters, including upwelling radiance, downwelling radiance, and transmissivity, is evaluated using simulated test datasets. Subsequently, a physical retrieval method that integrates spectral smoothness and the Hampel filter, referred to as the refined ISSTES algorithm, is implemented in the simulated data to assess its effectiveness.

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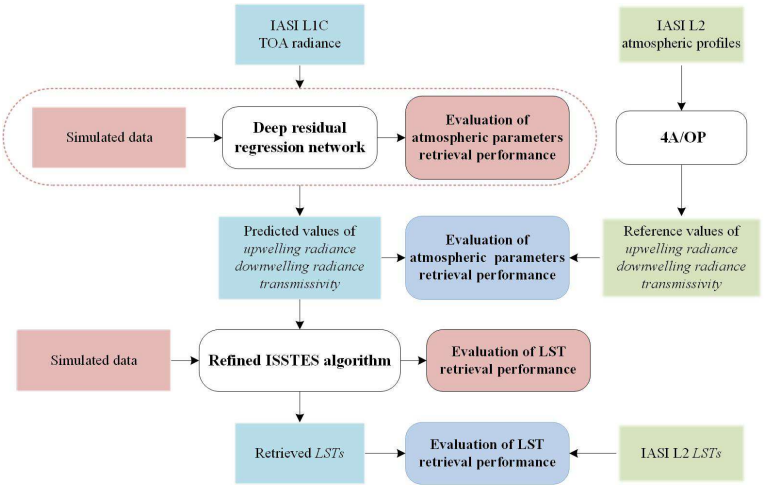


Fig. 1. Workflow of LST retrieval using the improved ResNet and ISSTES algorithm.

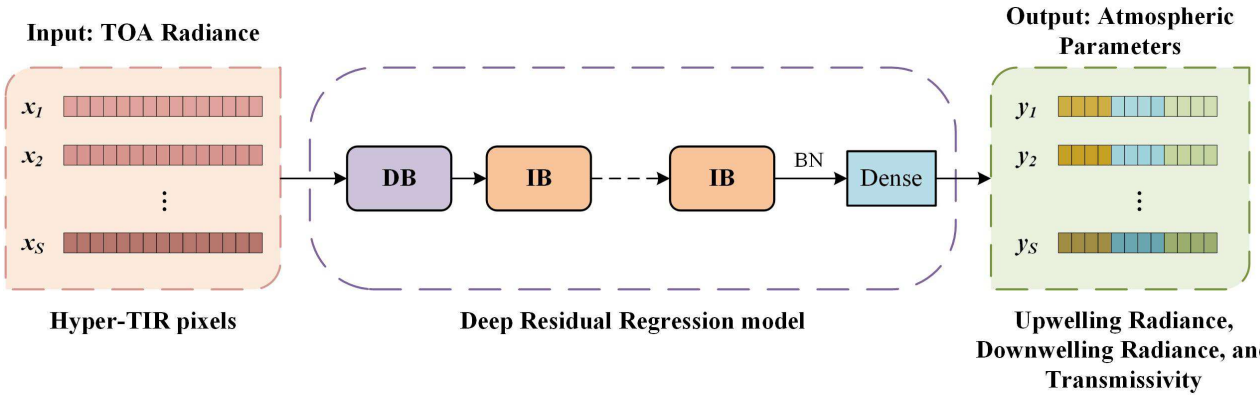


Fig. 2. Structure of the DR2N model used in this study. DB represents dense block, IB represents identity block, BN represents batch normalization, and dense denotes a fully connected layer. The number of IB is a tunable hyperparameter.

Finally, the proposed algorithm is applied to IASI L1C data for the study areas, and its consistency with IASI L2 products is also analyzed.

1) Deep Residual Regression Network: The universal approximation theorem asserts that neural networks are capable of approximating deterministic, nonlinear, continuous functions. Given the nonlinear nature of atmospheric radiative transfer, neural networks offer a viable approach for retrieving hyperspectral TIR atmospheric parameters. While deeper networks generally outperform shallower ones by more effectively approximating complex nonlinear functions, such as the radiative transfer function [27], [28], increasing the network depth can sometimes result in reduced accuracy due to issues such as vanishing gradients. To address this challenge, He et al. [29] introduced deep residual networks. Residual network (ResNet) is composed of multiple residual blocks, with the output of each block characterized by the fundamental relation

$$F(x) = H(x) - x \tag{4}$$

where F represents residual mapping learned by the network, x denotes the input data, and H refers to the original mapping between x and the feature that the network aims to learn. Based on the equation above, the original mapping is reformulated into $H(x) = F(x) + x$. Instead of directly modeling the $x \rightarrow H(x)$ mapping, this modified network learns the residuals between the two. Due to its excellent performance and adaptability in tasks such as image classification and object detection, ResNet has become one of the mainstream deep learning architectures.

The input to a nonlinear function, such as a radiative transfer function, is typically represented as a 1-D vector. However, the localized convolutional kernels in the original ResNet architecture have limited capacity to influence the entire sequence. As a result, the original ResNet architecture is suboptimal for addressing nonlinear regression problems [30]. Building on ResNet's structure, Chen et al. [30] developed a deep residual neural network specifically designed for nonlinear function regression. In this modified network, the residual blocks replace the convolutional and pooling layers with fully connected (dense) layers while retaining the batch normalization layers from the original model. These batch normalization layers serve as regularizers in certain cases, thereby eliminating the need for dropout and enabling the use of higher learning rates with reduced sensitivity to initialization. The architecture of DR2N proposed in this study is depicted in Figs. 2 and 3. The network dynamically selects between two processing blocks based on dimensional requirements:

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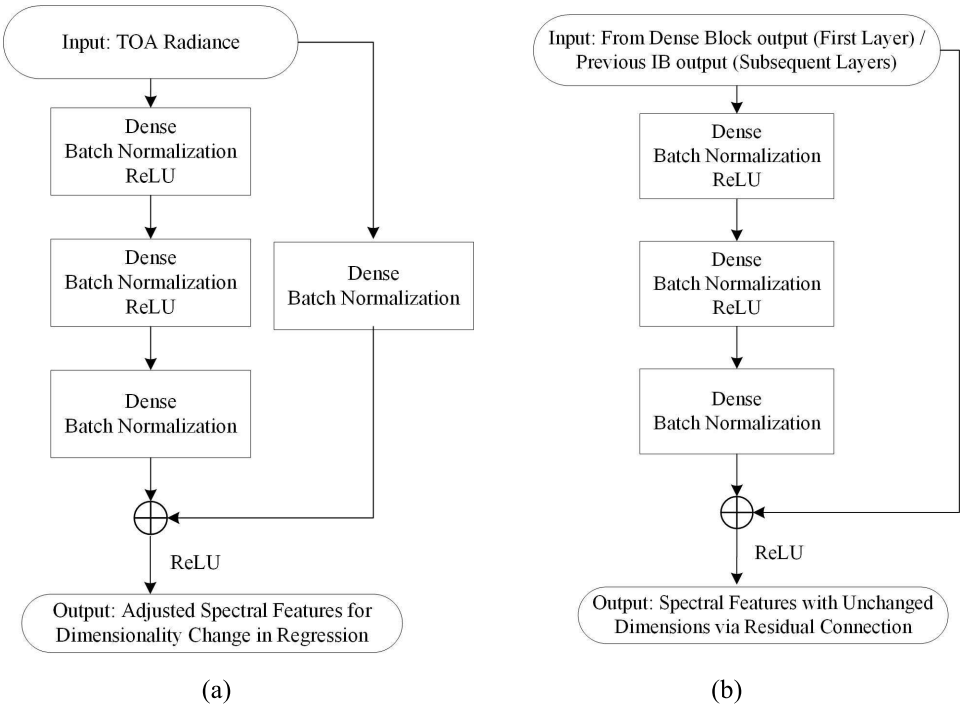


Fig. 3. Structures of residual blocks of DR2N: (a) dense block and (b) identity block.

the identity block for dimension-preserving operations and the dense block for size-altering transformations. The DR2N model consists of a combination of one dense block followed by multiple identity blocks, as illustrated in Fig. 2. The model inputs are TOA radiance, represented as spectral vectors of hyperspectral TIR pixels, while the outputs are parameters associated with upwelling radiance, downwelling radiance, and transmissivity. To address performance degradation and enhance computational efficiency, a dimensionality reduction technique is applied, which will be discussed in the data preprocessing of Section IV.

2) Refined Iterative Spectrally Smooth Temperature and Emissivity Separation: ISSTES is designed to retrieve LST and LSE from hyperspectral TIR radiance [7]. Based on (1), following the atmospheric correction, the surface-leaving radiance $L_{s,i}$ can be estimated using the relationship

$$L_{s,i} = \frac{L_i - L_{\mathrm{up},i}}{\tau_i}. \tag{5}$$

Based on (2), ε_i is derived as a function of T_s

$$\varepsilon_i = \frac{L_{s,i} - L_{\mathrm{down},i}}{B_i(T_s) - L_{\mathrm{down},i}}. \tag{6}$$

As indicated by (6), the inaccurate estimation of T_s causes the LSE spectrum to manifest pronounced convexities or concavities, a spectral artifact predominantly caused by atmospheric gas absorption [31]. Capitalizing on the inherent spectral smoothness of terrestrial LSE compared to atmospheric spectral features, the ISSTES algorithm establishes an iterative framework for simultaneous retrieval of LST and LSEs through hyperspectral TIR observations. The retrieval process is initiated by computing the initial LSE from the mean surface-leaving brightness temperatures (BTs), written as T_0. Then, an iterative refinement process is applied to optimize LST through minimization of the physically constrained cost function F_{loss}

$$F_{\mathrm{loss}} = \sum_{i=1}^{N} \left(L'_{s,i} - L_{s,i}\right)^2 \tag{7}$$

$$L'_{s,i} = \varepsilon_{\mathrm{SM},i} B_i(T_0) + \left(1 - \varepsilon_{\mathrm{SM},i}\right) L_{\mathrm{down},i}$$

$$\varepsilon_{\mathrm{SM},i} = \frac{\varepsilon_{i-1} + \varepsilon_i + \varepsilon_{i+1}}{3} \tag{8}$$

where ε_{i-1}, ε_i, and ε_{i+1} are three neighboring TIR channels; $\varepsilon_{\mathrm{SM},i}$ is the smoothed emissivity at channel i; and $L'_{s,i}$ denotes the iteratively updated surface-leaving radiance estimate using $\varepsilon_{\mathrm{SM},i}$ at each optimization step.

Subsequently, the optimal LST estimate emerges from minimizing the cost function F_{loss} via iterative T_0 adjustments. In addition, a constraint on the emissivity values, ensuring that they remain within the physical range of 0–1, is applied.

The accuracy of the ISSTES algorithm exhibits significant sensitivity to the degree of emissivity spectrum smoothness. As stated in (6), the smoothness of the emissivity spectrum is influenced by the surface-leaving radiance spectrum. Therefore, when the surface-leaving radiance spectrum exhibits irregularities, appropriate filtering techniques could be applied to smooth it, thereby improving the retrieval accuracy of the ISSTES algorithm in retrieving LST. The Hampel filter is a robust statistical method commonly used for outlier detection and removal in time series data or signal processing. It is based on the median absolute deviation (MAD), a robust variability estimator that outperforms standard deviation in outlier tolerance [32]. In view of this, the Hampel filtering technique is applied to smooth the surface-leaving radiance. The corresponding dataset is segmented into a sliding window

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of fixed size, and within each sliding window, the median is computed, followed by the calculation of absolute deviations from this median. The MAD is subsequently employed to define a threshold for identifying outliers. If a sample within the window deviates from the median by more than a specified multiple of the MAD, it is flagged as an outlier. These outliers can either be replaced with the median value of the window or removed entirely.

Mathematically, for a sample point $L_{s,i}$ in the surface-leaving radiance set L_s, it is considered an outlier if

$$|L_{s,i} - \text{median}\,(L_s)| > \delta \cdot \text{MAD} \tag{9}$$

where δ is a tunable parameter that controls the sensitivity of the filter, named threshold multiplier, and MAD is calculated as

$$\text{MAD} = \text{median}\left(|L_{s,i} - \text{median}\,(L_s)|\right). \tag{10}$$

After the Hampel filtering process detects and removes outliers, the filtered surface-leaving radiance values $L_{s,t}$ are determined by replacing any outlier values with the median value of the corresponding sliding window. Specifically, the postfiltering values of $L_{s,t}$ can be expressed mathematically as

$$L_{s,i}^{\text{filtered}} = \begin{cases} L_{s,i}, \text{if}|L_{s,i} - \text{median}\,(Ls)| \leq \delta \cdot \text{MAD} \\ \text{median}\,(L_s), \text{if}|L_{s,i} - \text{median}\,(Ls)| > \delta \cdot \text{MAD} \end{cases} \tag{11}$$

where $L_{s,i}^{\text{filtered}}$ denotes the postfiltering value of the surface-leaving radiance.

This filtering process ensures that those values deemed outliers are replaced with the median, thereby improving the spectral smoothness of the radiance data. Thus, after applying the Hampel filter, the surface-leaving radiance values $L_{s,t}$ are smoothed for LST retrieval.

III. Dataset

A. Simulated Data

In this study, the thermodynamic initial guess retrieval (TIGR) atmospheric profile database (TIGR_2000_v1.2_43lev), developed by the Laboratoire de Météorologie Dynamique (LMD), is utilized for the simulation of hyperspectral TIR data considering different atmospheric conditions and surface varieties. The TIGR database includes over 80 000 global soundings, providing 2311 representative profiles covering a wide range of Earth's atmospheric types [33]. It defines a 43-layer atmospheric structure, with each layer providing data on pressure, temperature, H_2O concentration, and O_3 density. Atmospheric profiles containing at least one layer with relative humidity (RH) >90%, or two consecutive layers with RH >85%, are identified as cloudy conditions and subsequently removed from the simulation dataset [34]. Ultimately, 1276 cloud-free atmospheric profiles are selected for simulation in this study. As shown in Figs. 4 and 5, the selected profiles have the bottom atmospheric temperature (T_{bottom}) ranging from 231.25 to 314.16 K, and total precipitable water (TPW) spans 0.05–7.75 g/cm^2.

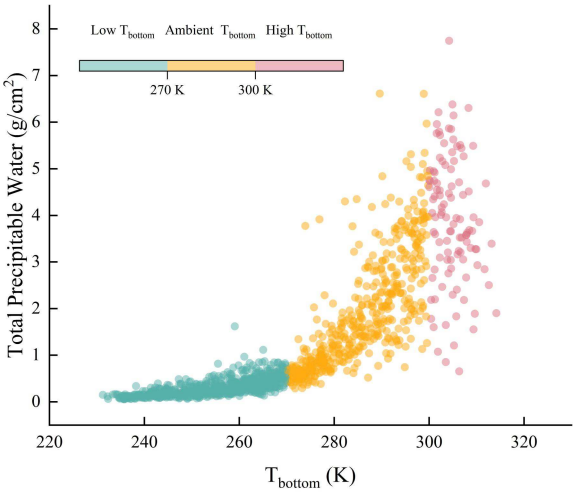


Fig. 4. Distribution of bottom atmospheric temperature versus TPW for the 1276 cloud-free profiles in the TIGR database. The three colors represent low-temperature, ambient-temperature, and high-temperature atmospheric profiles, respectively.

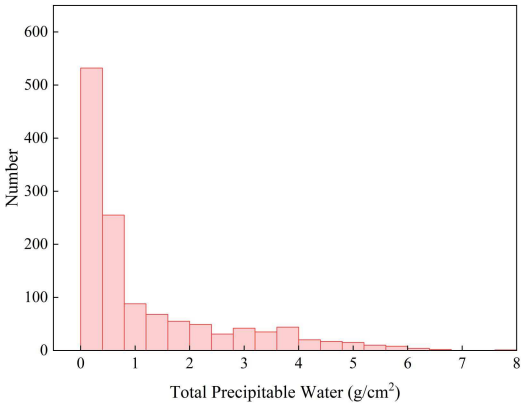


Fig. 5. Histogram of TPW of 1276 cloud-free profiles in the TIGR database.

Subsequently, the corresponding atmospheric upwelling radiance, downwelling radiance, and transmissivity are calculated using 4A/OP model, a specialized software designed for hyperspectral TIR radiative transfer, which is developed, maintained, and validated by LMD. The 4A model is a line-by-line radiative transfer model that enables rapid, accurate computations using a pregenerated optical thickness database, Atlas. The 4A model generates spectral radiance outputs covering the 600–3000-cm^{-1} range, achieving a spectral resolution as fine as 0.0005 cm^{-1} [35]. In this study, the viewing zenith angle (VZA) for all simulated data is set to 0°, and the emissivity spectra are selected from the ASTER spectral library, which provides a comprehensive collection of spectra for natural and artificial objects. A total of 56 emissivity spectra are selected, as illustrated in Fig. 6, including four for water bodies, three for frost and snow, four for plantations, 18 for rocks, 25 for soils, and two for man-made features. The emissivity covers the 645–1600-cm^{-1} interval at 0.25 cm^{-1} spectral resolution, ensuring minimal contamination from solar components. Moreover, to enhance simulation representativeness, the LST is acquired from T_{bottom} extracted from profiles, varying from −6 to + 4 K in steps of 2 K for $T_{\text{bottom}} \leq 270$ K,

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TABLE I
PARAMETERS OF MAIN SPECTRAL REGIONS USED BY THE IASI INSTRUMENT

Name	Spectral region/cm^{-1}	Absorption band	IASI application	NEΔT /K@280K
R1	650-770	CO_2	Temperature profile	0.20
R2	790-980	Atmospheric window	Surface and cloud properties	0.24
R3	1000-1070	O_3	O3 sounding	0.20
R4	1080-1150	Atmospheric window	Surface and cloud properties	0.24
R5	1210-1650	H_2O	Humidity profile; CH4 and N2O column amount	0.20
R6	2100-2150	CO	CO column amount	0.36
R7	2150-2250	N_2O and CO_2	Temperature profile N2O column amount	0.36
R8	2350-2420	CO_2	Temperature profile	0.36
R9	2420-2700	Atmospheric window	Surface and cloud properties	0.36
R10	2700-2760	CH_4	CH_4 column amount	1.53

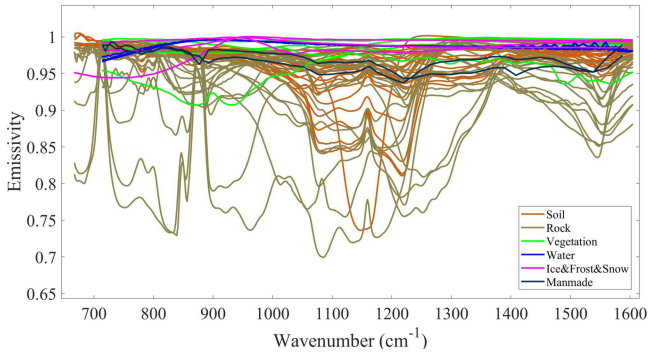


Fig. 6. Selected emissivity spectra for simulation.

from −5 to + 5 K in steps of 2 K for 270 K < $T_{\text{bottom}} \leq$ 300 K, and from −4 to + 6 K in steps of 2 K for $T_{\text{bottom}} >$ 300 K. According to (1) and (2), the TOA radiances for various surfaces and atmospheric conditions are generated.

B. Iasi Satellite Data

The infrared atmospheric sound interferometer (IASI) serves as a key instrument integrated into the MetOp satellite series, codeveloped by the European Space Agency (ESA) and the French National Center for Space Studies (CNES). Operating as a Fourier transform spectrometer, IASI performs observations of Earth's infrared radiative emissions. The instrument operates across the 645–2760-cm^{-1} (3.62–15.5-μm) spectral band, with 12-km spatial resolution and 0.25-cm^{-1} spectral resolution. The key parameters of the main spectral regions utilized by the IASI instrument are summarized in Table I [36], [37].

The IASI L1C product contains infrared radiance spectra, with each sounding pixel in the L1C data encompassing 8461 spectral samples. The IASI Level 2 (L2) product comprises a series of retrieval outputs derived by applying algorithms on the L1C data, incorporating numerical weather prediction (NWP) models and other satellite observations, including AVHRR and AMSU-A. These retrievals provide a range of atmospheric data, such as atmospheric temperature profiles, humidity profiles, ozone profiles, surface temperature, surface emissivity, cloud parameters, and additional data [38], [39].

To assess the accuracy of the hyperspectral TIR LST retrieval algorithm proposed, four study areas, Spain, North Africa, Hulunbuir, and the Yellow Sea, are selected. These regions cover a variety of typical surface types and exhibit high clear-sky rates. The land cover map for the four regions, extracted from the "Land Cover of the World in the Year 2000" dataset provided by the European Commission Joint Research Centre [40], is shown within the red rectangles in Fig. 7. The eastern part of Spain features a temperate Mediterranean climate, characterized by plateaus interspersed with mountains, forests, and other vegetation. The main land covers in this area include cultivated and managed areas, tree cover, shrub cover, artificial surfaces, and associated areas. North Africa exhibits a tropical desert climate, dominated by plateaus and deserts. The Hulunbuir grassland region features a mesothermal continental monsoon climate and is classified within the semiarid zone. Its western part represents a typical temperate grassland, with terrain primarily consisting of low hills and river valley plains. The Yellow Sea region lies within a temperate monsoon climate zone and experiences high clear-sky rates during the fall and winter.

In this study, both the IASI L1C and IASI L2 datasets for 2023 are used, with 74 clear-sky observations for Spain and North Africa, 50 for Hulunbuir, and 44 for the Yellow Sea, as shown in Table II. Since the DR2N model is trained by data under nadir viewing and cloud-free conditions, only IASI pixels with VZA below 10° and no cloud coverage are selected to ensure the model's adaptability. The corresponding TPW and VZA distributions for the selected pixels are shown in Fig. 8. It is observed that the TPW values of the selected IASI samples range between 0.2 and 2.2 g/cm^2, which aligns with the distribution of over 50% of the TIGR profiles. In addition, the VZA values of the selected pixels fall between 1° and 9°.

IV. RESULTS AND DISCUSSION

A. Data Preprocessing

Prior to training the DR2N model, data standardization and principal component analysis (PCA) are employed to enhance model convergence and computational efficiency. These preprocessing steps ensure that input features contribute equally to the learning process while reducing data redundancy.

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TABLE II
INFORMATION ON IASI CLEAR-SKY OBSERVATIONS FOR FOUR RESEARCH AREAS

Area	Sensing Start Time (UTC)	Geographical Scope (°) (minLon, minLat, maxLon, maxLon)	Sample Number
Spain	2023.5.1 19:47	-4.5, 39, -2.5, 41	32
North Africa	2023.5.1 19:47	0, 28, 2, 30	40
Hulunbuir	2023.8.18 2:00	118, 46.2, 120, 50.2	52
The Yellow Sea	2023.2.19 1:32	121, 33, 125.5, 36	44

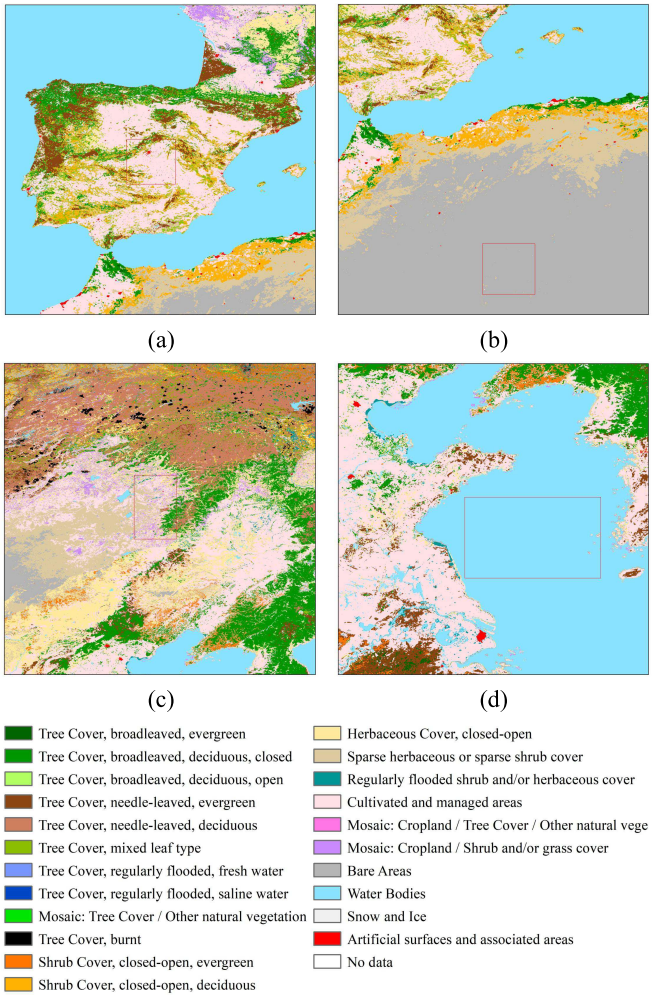


Fig. 7. Surface covers in study areas (a) Spain, (b) North Africa, (c) Hulunbuir, and (d) Yellow Sea with 1-km resolution extracted from the GLC 2000 dataset.

The first step involves standardizing both input and output data. Data standardization mitigates the undesirable effects caused by singular sample data and ensures that all features contribute equally to the analysis. Specifically, for each feature vector of the simulated hyperspectral TIR TOA radiance, upwelling radiance, downwelling radiance, and transmissivity, the mean value is subtracted, and the result is divided by the standard deviation. In addition, radiative values, including TOA radiance, upwelling radiance, and downwelling radiance, are converted into BT.

After data standardization, although hyperspectral TIR data provide extensive spectral information regarding atmospheric and surface parameters, they also exhibit substantial redundancy, which increases computational costs and reduces efficiency. Therefore, PCA is further employed to reduce the dimensionality of three atmospheric parameters (upwelling radiance, downwelling radiance, and transmissivity). To preserve essential characteristics while reducing complexity, varying numbers of principal components, ranging from 1 to 150, are extracted. The final number is determined based on reconstruction error. As shown in Fig. 9, the explained variances for upwelling radiance, downwelling radiance, and transmissivity exceed 99% when 50 principal components are retained. Simultaneously, the reconstruction error stabilizes, achieving RMSEs of 0.054 K, 0.021 K, and 2.6×10^{-4} for upwelling radiance, downwelling radiance, and transmissivity, respectively.

Subsequently, the simulated dataset is divided into two subsets: one for model training with 80% of the dataset and the other allocated to testing purposes. To ensure the representativeness of the dataset, both subsets are randomly sampled according to the specified ratio.

B. Hyperparameter Settings in DR2N

Hyperparameters are predefined parameters in machine learning and deep learning models that influence performance and behavior. Common hyperparameters include batch size, maximum epoch number, learning rate, loss function, optimization algorithm, number of hidden layers, units, and activation function. In this study, the batch size is fixed at 512, with a maximum of 200 training epochs allowed. The adaptive moment estimation (Adam) optimizer, initialized with a learning rate of 0.01, is employed alongside the mean square error (mse) loss function. The Adam optimizer is a widely used gradient descent strategy in deep learning, designed to dynamically adjust the learning rate for each parameter. In addition, the training process is monitored using early stopping, a method employed to mitigate overfitting by terminating training when the model's validation performance ceases to improve [41]. Specifically, the patience threshold for early stopping is established at 20 epochs.

Based on the aforementioned settings, the DR2N model is trained multiple times with varying widths to determine the optimal configuration. The results indicate that the testing loss is minimized at a width of 200. Therefore, as a general rule, the width of each hidden layer is fixed at 200. Moreover,

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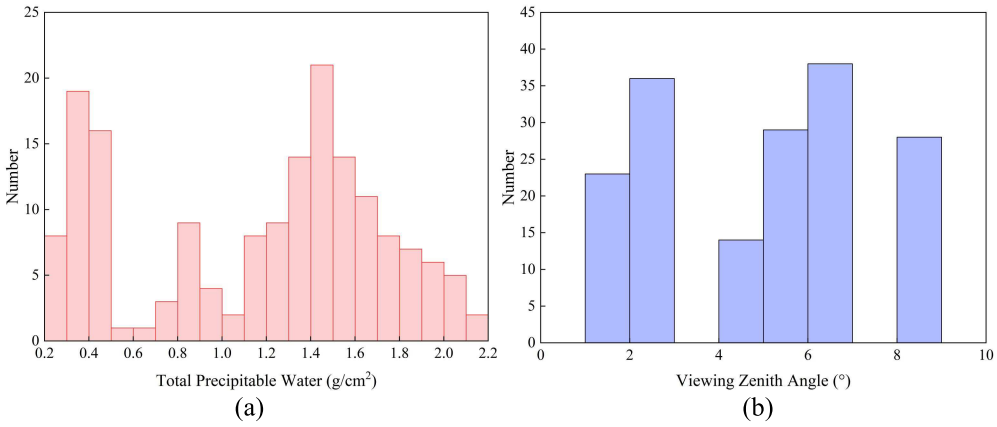


Fig. 8. Histograms of sample distribution of (a) TPW and (b) VZA.

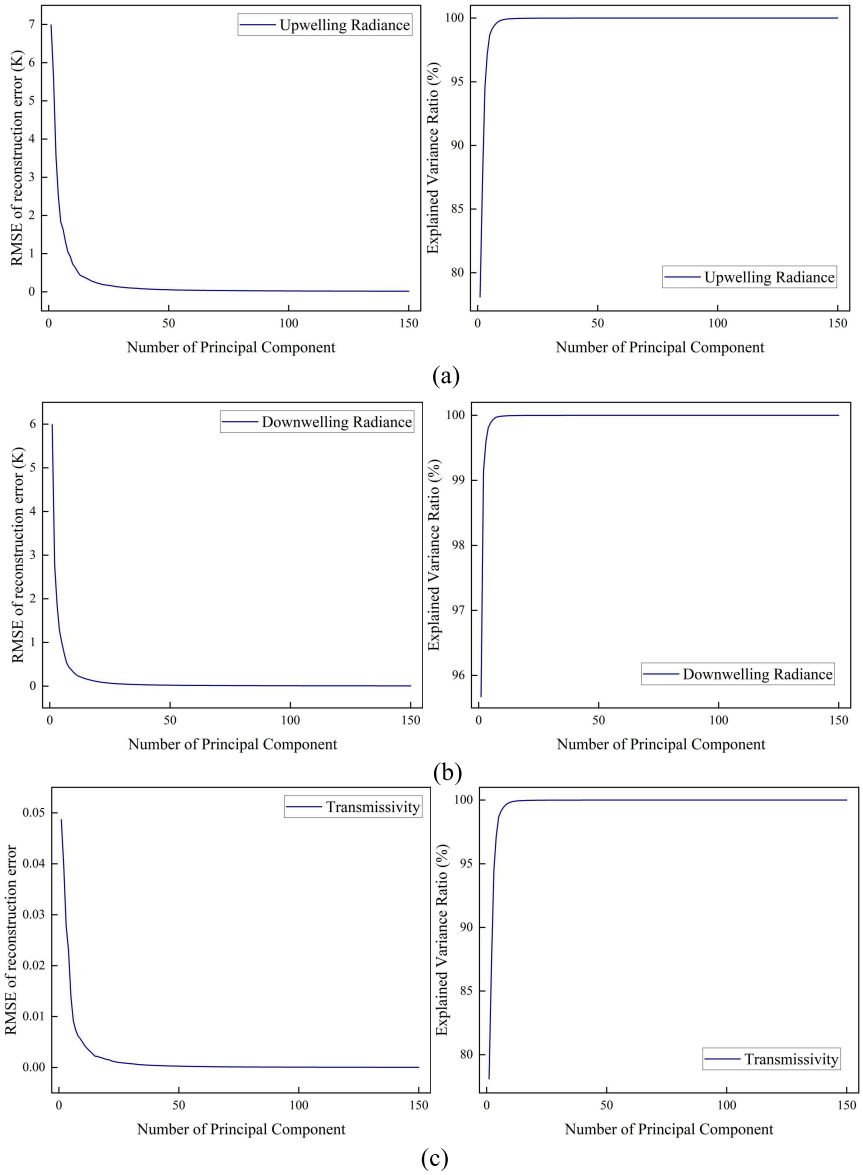


Fig. 9. Reconstruction error and explained variance ratio change with the number of principal components for: (a) upwelling radiance, (b) downwelling radiance, and (c) transmissivity.

since model parameter initialization is random, the performance of each model is averaged over five training runs with the same hyperparameters to ensure reliable results, particularly for models with varying depths. The retrieval RMSEs

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TABLE III
PERFORMANCE OF DR2N ON THE TEST DATASET

Width	Depth/Block Number	RMSE of upwelling radiance/K	RMSE of downwelling radiance/K	RMSE of transmissivity	Stopping epoch	Training time (H:MM)
200	4/1	0.295	0.419	0.0021	170	1:28
200	7/2	0.183	0.232	0.0012	124	1:13
200	10/3	0.146	0.180	0.0010	158	1:46
200	13/4	0.128	0.157	0.0008	181	2:21
200	**16/5**	**0.126**	**0.155**	**0.0008**	**159**	**2:24**
200	19/6	0.118	0.145	0.0007	156	2:36

Note: the boldface text represents the DR2N model selected in this study.

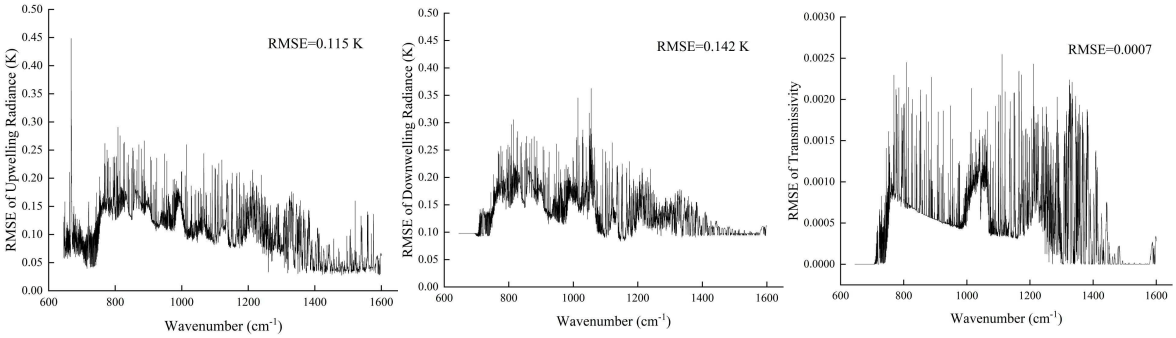


Fig. 10. Retrieval error ($RMSE_i$) on the test dataset for (a) upwelling radiance, (b) downwelling radiance, and (c) transmissivity.

of atmospheric parameters are calculated by the following equations:

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{j=1}^{m}\sum_{i=\lambda_0}^{\lambda}\left(x_{\mathrm{pre},i,j} - x_{\mathrm{ref},i,j}\right)^2}{n \cdot m}} \tag{12}$$

$$\mathrm{RMSE}_i = \sqrt{\frac{\sum_{j=1}^{m}\left(x_{\mathrm{pre},i,j} - x_{\mathrm{ref},i,j}\right)^2}{m}} \tag{13}$$

where RMSE represents the total retrieval RMSE of atmospheric transmissivity, upwelling, or downwelling radiance for a set of samples over a spectral range, n denotes the total number of channels, and m denotes the total number of samples. i denotes a specific channel within the spectral interval from λ_0 to λ cm^{-1}, j is the ordinal number of the sample, x_{pre} is the retrieved atmospheric transmissivity or upwelling/downwelling radiance, and x_{ref} is the true value of atmospheric transmissivity or upwelling/downwelling radiance. RMSE_i is the retrieval RMSE of atmospheric transmissivity, upwelling, or downwelling radiance for a set of samples at channel i.

Table III presents the performance of DR2N models with different depths on the test dataset. Notably, with a fixed width of 200, the RMSE of the model for the three parameters progressively decreases as the model depth increases from 4 to 19. Specifically, RMSE for upwelling radiance decreases from 0.295 to 0.118 K, for downwelling radiance from 0.419 to 0.145 K, and for transmissivity from 0.0021 to 0.0007. At depths of 13, 16, and 19, the accuracy of the model on the test dataset converges, with the depth of 16 requiring fewer training epochs. To balance model accuracy and training stability, the DR2N model with a width of 200 and a depth of 16 is selected.

C. Algorithm Performance on Simulated Data

1) Atmospheric Parameters' Retrieval From Simulated Data: In this study, the performance of DR2N on simulated test data is initially investigated. Fig. 10 illustrates the retrieval errors of various parameters at each channel on the test dataset. The RMSEs for upwelling radiance, downwelling radiance, and transmissivity across 645–1600 cm^{-1} are 0.115 K, 0.142 K, and 7×10^{-4}, respectively. For any given channel, RMSE_i for these three parameters does not exceed 0.45 K, 0.37 K, and 0.0026, respectively. As shown in Fig. 10(a), the reconstruction errors for upwelling radiance are relatively higher in the range of 740–1300 cm^{-1}; in Fig. 10(c), channels with significant variations in the TOA radiance spectrum exhibit larger RMSE fluctuations for transmissivity compared to other channels. Similar trends are also observed for the RMSEs of upwelling and downwelling radiances. In addition, the RMSEs for transmissivity in atmospheric window channels with high transmissivity are relatively larger than those in other regions, while a significant peak is observed in the ozone absorption channels. These findings indicate that the model's ability to retrieve atmospheric parameters varies across different channels, closely correlating with the atmospheric absorption spectrum, particularly for transmissivity. Moreover, the RMSEs of the three parameters are influenced not only by the model retrieval error but also by errors introduced during the principal component transformation.

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TABLE IV
RMSE AND R^2 BETWEEN RETRIEVED AND REFERENCE ATMOSPHERIC PARAMETERS ACROSS ALL CHANNELS FOR SAMPLES AT FOUR RESEARCH AREAS

Area	RMSE of upwelling radiance/K	RMSE of downwelling radiance/K	RMSE of transmissivity	R^2 of upwelling radiance	R^2 of downwelling radiance	R^2 of transmissivity
Spain	**3.27**	8.43	**0.020**	0.986	0.948	0.997
North Africa	**2.41**	6.92	**0.012**	0.990	0.964	0.999
Hulunbuir	**3.37**	**4.30**	0.026	0.973	0.978	0.994
The Yellow Sea	**3.43**	**3.92**	0.021	0.990	0.991	0.997
Total	**3.14**	5.61	0.020	0.984	0.972	0.997

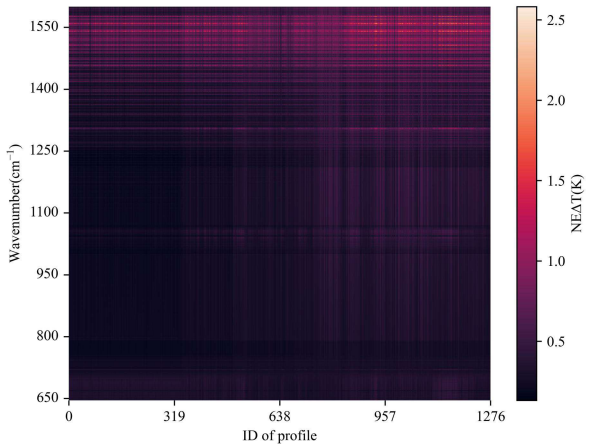


Fig. 11. NEΔT distribution of simulated data.

Meanwhile, to gauge the model's sensitivity to noise, the DR2N model is also applied to simulated data with added noise. For each channel, the noise level of the instrument is computed, as shown in Fig. 11. Notably, the NEΔT of the simulated data ranges from 0.1 to 2.6 K and increases with wavenumber.

Furthermore, random noise is introduced across various channels under each atmospheric profile, characterized by a mean of 0 and a standard deviation equivalent to NEΔT. The simulated hyperspectral TIR TOA radiance with added noise is then input into the DR2N model to retrieve the noise-added upwelling radiance, downwelling radiance, and transmissivity, and the model's RMSE on the noise-added test dataset is shown in Fig. 12. Notably, compared to the output without noise (see Fig. 10), the RMSE of upwelling radiance increases from 0.115 to 0.134 K, the RMSE of downwelling radiance increases from 0.142 to 0.164 K, and the RMSE of transmissivity increases from 7×10^{-4} to 8×10^{-4}. However, the trends and distributions of the RMSEs for the three atmospheric parameters in the 645–1600-cm^{-1} range remain nearly identical in the noise-added test dataset. These results reveal that the proposed DR2N is robust and relatively insensitive to noise.

2) LST Retrieval From Simulated Data: To assess the accuracy of the proposed refined ISSTES algorithm, simulated data from three typical surface types, bare land, sea water, and grassland, are used. These surface types are chosen to represent diverse emissivity characteristics and atmospheric interactions, providing a comprehensive assessment of the algorithm's robustness. As depicted in Fig. 13, the RMSE and bias between the retrieved and reference LSTs for the overall simulated dataset are 1.89 and −0.19 K, respectively. Breaking down the results by surface type, the RMSE values for bare land, sea water, and grassland are 1.40, 1.47, and 2.56 K, respectively, with corresponding biases of 0.60, 0.93, and −2.09 K. Notably, grassland exhibits the highest LST retrieval errors among the three surface types. This discrepancy is likely due to the presence of mixed pixels commonly found in grassland areas. Such heterogeneity can introduce significant challenges in accurately representing the true LST, as variations in vegetation cover, soil exposure, and moisture content can lead to spectral inconsistencies. These factors complicate the atmospheric correction and emissivity separation processes, thereby resulting in larger retrieval errors. Overall, the results show that the refined ISSTES algorithm performs satisfactorily for bare land and sea water, with relatively low errors. However, the higher error for grassland highlights the need for further refinement of the algorithm or additional preprocessing steps, particularly for heterogeneous surface types. This insight provides a valuable direction for improving the algorithm's applicability across more complex surface conditions.

D. Algorithm Performance on Iasi Observation

1) Atmospheric Parameters' Retrieval From Iasi Observation: To evaluate the accuracy of the trained DR2N model, IASI L1C data are used to predict upwelling radiance, downwelling radiance, and transmissivity for atmospheric correction. Simultaneously, the IASI L2 atmospheric profile products, including temperature profiles, humidity profiles, and ozone profiles, are also input into the 4A/OP model to generate reference values for the three atmospheric parameters. By comparing the predicted values from DR2N with the reference values from 4A/OP, the RMSEs for each parameter are calculated over four study sites in two spectral ranges, namely, 800–980 and 645–1600 cm^{-1} (all channels). These spectral ranges are selected to assess the model's performance under different atmospheric absorption conditions, thereby providing a comprehensive evaluation of its retrieval accuracy. The RMSE for each atmospheric parameter in each study area is calculated using (12).

As shown in Table IV, at the four research areas, the overall RMSEs across all channels for upwelling radiance,

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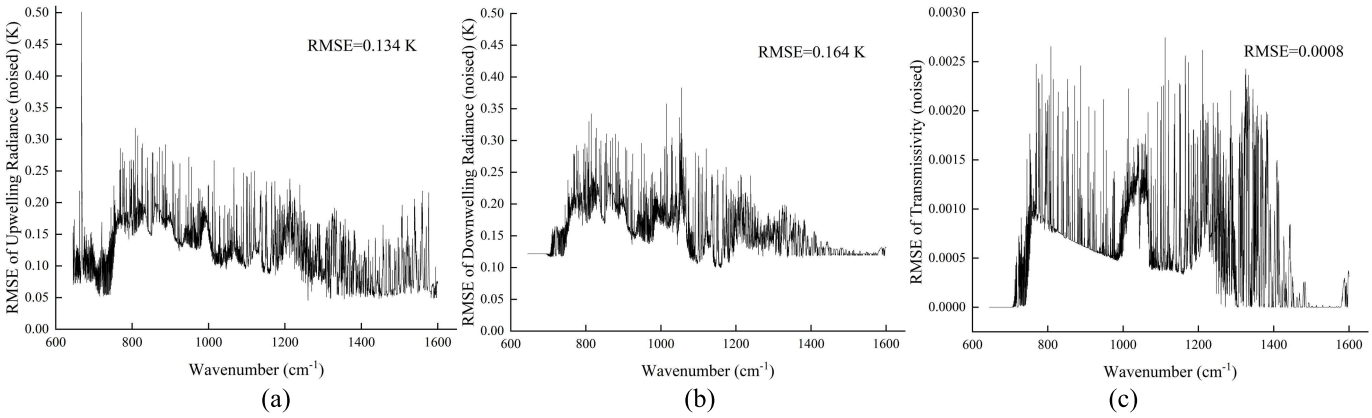


Fig. 12. Retrieval error ($RMSE_i$) of the noised test dataset for (a) upwelling radiance, (b) downwelling radiance, and (c) transmissivity.

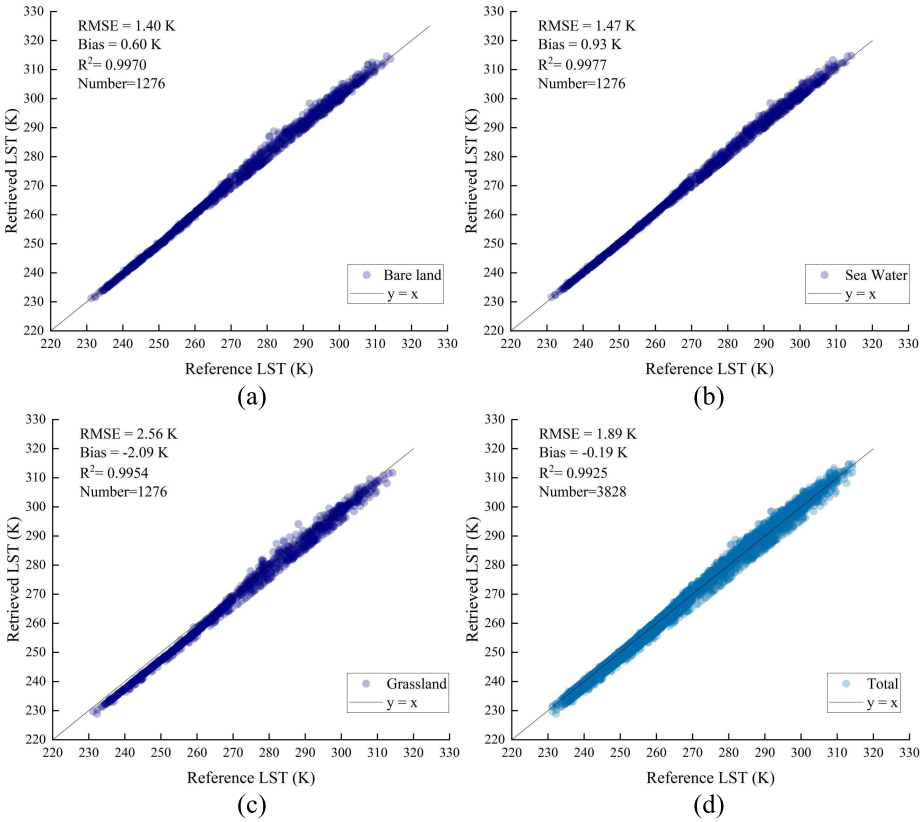


Fig. 13. Scatterplots of retrieved LST versus reference LST for simulated data (a) bare land, (b) sea water, (c) grassland, and (d) all three surface types combined.

downwelling radiance, and transmissivity are 3.14 K, 5.61 K, and 0.020, respectively, and North Africa achieves the highest accuracy for acquiring upwelling radiance and transmissivity, with RMSE of 2.41 K and 0.012; in contrast, the Yellow Sea exhibits the lowest accuracy for these two parameters, with RMSE of 3.43 K and 0.021, respectively. However, for downwelling radiance, the lowest RMSE is observed in the Yellow Sea, while the highest is in Spain, showing a distinct distribution pattern compared to upwelling radiance and transmissivity.

Specifically, we also investigate the atmospheric parameters' retrieval difference for the spectral range used in LST retrieval, namely, 800–980-cm^{-1} range (see Table V), and it is noted the overall RMSE for upwelling radiance increases to 4.55 K, which is higher than the RMSE observed across all channels, whereas RMSEs for downwelling radiance and transmissivity decrease slightly to 5.18 K and 0.020, respectively. Furthermore, in North Africa, atmospheric correction maintains the highest accuracy for the three parameters, with RMSE of 3.15 K, 3.98 K, and 0.015, respectively. In contrast, the Yellow Sea exhibits the lowest accuracy for upwelling and downwelling radiances, with RMSE of 5.37 and 5.94 K, respectively, while transmissivity shows the lowest RMSE in North Africa and the highest in Hulunbuir. It is worth noting that North Africa

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TABLE V

RMSE AND R^2 BETWEEN RETRIEVED AND REFERENCE ATMOSPHERIC PARAMETERS ACROSS 800–980 CM^{-1} FOR SAMPLES AT FOUR RESEARCH AREAS

Area	RMSE of upwelling radiance/K	RMSE of downwelling radiance/K	RMSE of transmissivity	R^2 of upwelling radiance	R^2 of downwelling radiance	R^2 of transmissivity
Spain	5.02	**5.85**	0.021	0.870	0.843	0.951
North Africa	3.15	**3.98**	0.015	0.957	0.942	0.986
Hulunbuir	4.64	5.05	**0.024**	0.835	0.817	0.944
The Yellow Sea	5.37	5.94	**0.018**	0.815	0.805	0.899
Total	4.55	**5.18**	**0.020**	0.865	0.848	0.943

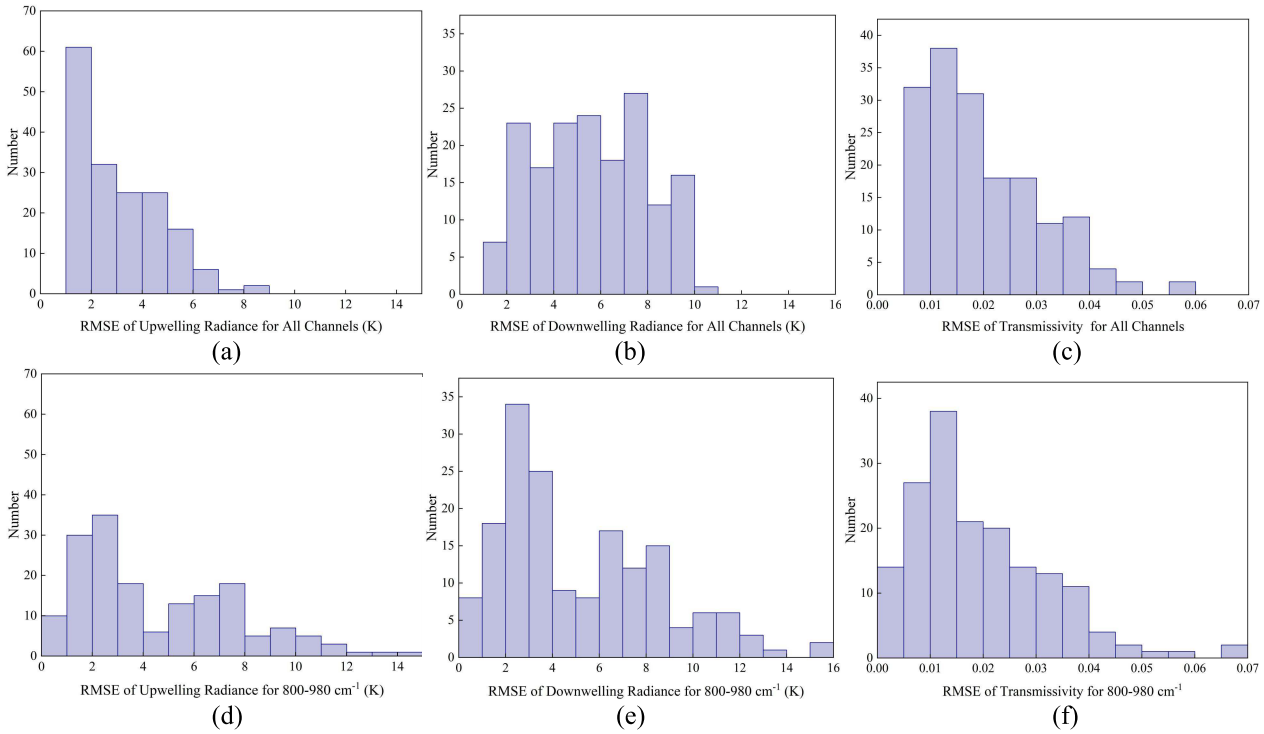


Fig. 14. Histograms of retrieval difference ($RMSE$) distributions for three atmospheric parameters across the 800–980-cm^{-1} spectral range and all channels based on samples from four research areas. For all channels: (a) upwelling radiance, (b) downwelling radiance, and (c) transmissivity. For the 800–980 cm^{-1} range: (d) upwelling radiance, (e) downwelling radiance, and (f) transmissivity.

demonstrates the best overall precision, likely attributed to its homogeneous and stable surface cover.

Generally, RMSE for downwelling radiance is higher than that for upwelling radiance. RMSEs for upwelling radiance are higher in the 800–980-cm^{-1} range than those across all channels; this would be attributed to the reduced amount of information available for radiance correction in the atmospheric window so that the atmospheric correction model captures fewer relevant atmospheric features, leading to lower correction accuracy in these channels. In addition, the coefficient of determination (R^2) in all channels at the four research areas is 0.984, 0.972, and 0.997 for upwelling radiance, downwelling radiance, and transmissivity, respectively, higher than those in the 800–980-cm^{-1} range (0.865, 0.848, and 0.943). This demonstrates that the DR2N model better captures atmospheric parameter trends across all channels, especially outside the 800–980-cm^{-1} range. The effect is particularly notable for upwelling radiance, which also shows lower RMSE, further indicating improved predictive performance. For downwelling radiance in Spain and North Africa, RMSEs are higher in the 800–980-cm^{-1} range than across all channels, whereas, in Hulunbuir and the Yellow Sea, RMSEs are lower in this range. For transmissivity, the retrieval accuracy shows less fluctuation between the 800 and 980 cm^{-1} and all channels. In Spain and North Africa, RMSEs are lower in the 800–980-cm^{-1} range than across all channels, while, in Hulunbuir and the Yellow Sea, RMSEs are higher in the 800–980-cm^{-1} range.

Furthermore, to evaluate the model's comprehensive performance on the spectra, the RMSE histogram distributions for all samples are analyzed and presented in Fig. 14, which represent the retrieval errors for three atmospheric parameters across the 800–980-cm^{-1} spectral range and all channels, calculated using (12) with $m = 1$, for individual samples from four research areas. It is demonstrated that the RMSEs for upwelling radiance range from 1 to 9 K across all channels and from 0.5 to 14.3 K within the 800–980-cm^{-1} range; for

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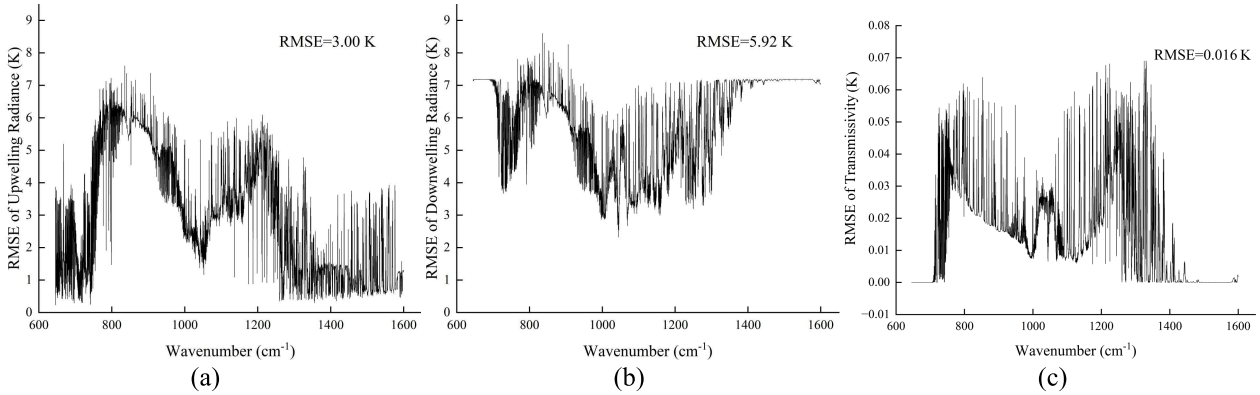


Fig. 15. Spectral distribution of retrieval difference ($RMSE_i$) for samples from four research areas: (a) upwelling radiance, (b) downwelling radiance, and (c) transmissivity.

TABLE VI
LST RETRIEVAL RESULTS FOR FOUR RESEARCH AREAS

Area	RMSE (K)	Bias (K)	R^2	Sample Number	Surface Type
Spain	0.81	-0.68	0.9701	32	Cultivated and managed land
North Africa	0.28	-0.06	0.9482	40	Bare land
Hulunbuir	0.45	-0.05	0.9766	52	Grassland, Cultivated and managed land
The Yellow Sea	0.44	-0.36	0.9801	44	Sea water
Total	0.51	-0.25	0.9966	168	-

downwelling radiance, RMSEs range from 1 to 11 K across all channels and from 0.4 to 15.8 K within the 800–980-cm^{-1} range; and for transmissivity, RMSEs range from 0.008 to 0.060 across all channels and from 0.002 to 0.070 within the 800–980-cm^{-1} range. In addition, compared to all channels, the RMSEs for the three parameters generally exhibit higher thresholds and greater dispersion within the 800–980-cm^{-1} range, resulting in the majority of RMSE values for these parameters at the four sites being higher in this range. Furthermore, the spectral distributions of RMSEs for all samples are also investigated. As shown in Fig. 15, the overall RMSEs across all channels for upwelling radiance, downwelling radiance, and transmissivity are 3.00 K, 5.92 K, and 0.016, respectively. Within the atmospheric window regions, such as 800–980 and 1150–1250 cm^{-1}, the RMSEs are relatively higher than that in other channels, particularly for upwelling radiance, which exhibits the most pronounced deterioration in accuracy within the 800–980-cm^{-1} range compared to the other two parameters.

In brief, the atmospheric correction results of the DR2N model indicate that compared to its performance on the simulated dataset test set, the retrieval accuracy of the three atmospheric parameters significantly decreases when applied to IASI observation data. In addition, the accuracy gap between atmospheric upwelling and downwelling radiances becomes more pronounced. This suggests that the trained atmospheric correction model lacks sufficient generalization capability, which would introduce errors to the LST retrieval. Therefore, in this study, the ISSTES is refined, in which the Hampel filter is used to reduce the influence of the atmospheric parameters' retrieval difference.

2) LST Retrieval From Iasi Observation: After performing atmospheric correction via the DR2N model, LST is derived from the surface-leaving radiance using the refined ISSTES algorithm. The spectral range of 800–980 cm^{-1} lies within the atmospheric window region, making it particularly suitable for acquiring surface characterization parameters. However, as mentioned earlier, if atmospheric correction is incomplete, residual atmospheric information, manifested as outliers in the surface-leaving radiance spectrum, can diminish the accuracy of LST retrieval. To address this issue, the surface-leaving radiance spectra are filtered using the Hampel filter before applying ISSTES to suppress the effect of outliers on subsequent calculations. A window size of 10 and a tunable parameter δ are used for the filtering.

Fig. 16 shows examples of comparison between surface-leaving radiance without and with Hampel filtering at the four study sites. It is observed that the surface-leaving radiance outliers exhibit a similar channel distribution, particularly near 813 and 853 cm^{-1}, where the maximum outliers are consistently observed at each site. This phenomenon is likely due to insufficient feature learning by the DR2N model for atmospheric parameters in these channels. Notably, the Hampel filter could effectively smooth the surface-leaving spectra, removing outliers at all four sites while preserving the original trend of the data.

Furthermore, given the challenges of obtaining sufficient large-scale ground measurements for validating satellite observations, the retrieved LST (denoted as IASI LST1) in this study are compared with the published IASI LST extracted from IASI L2 datasets (denoted as IASI LST2) to ensure spatial and temporal consistency. The scatterplot between

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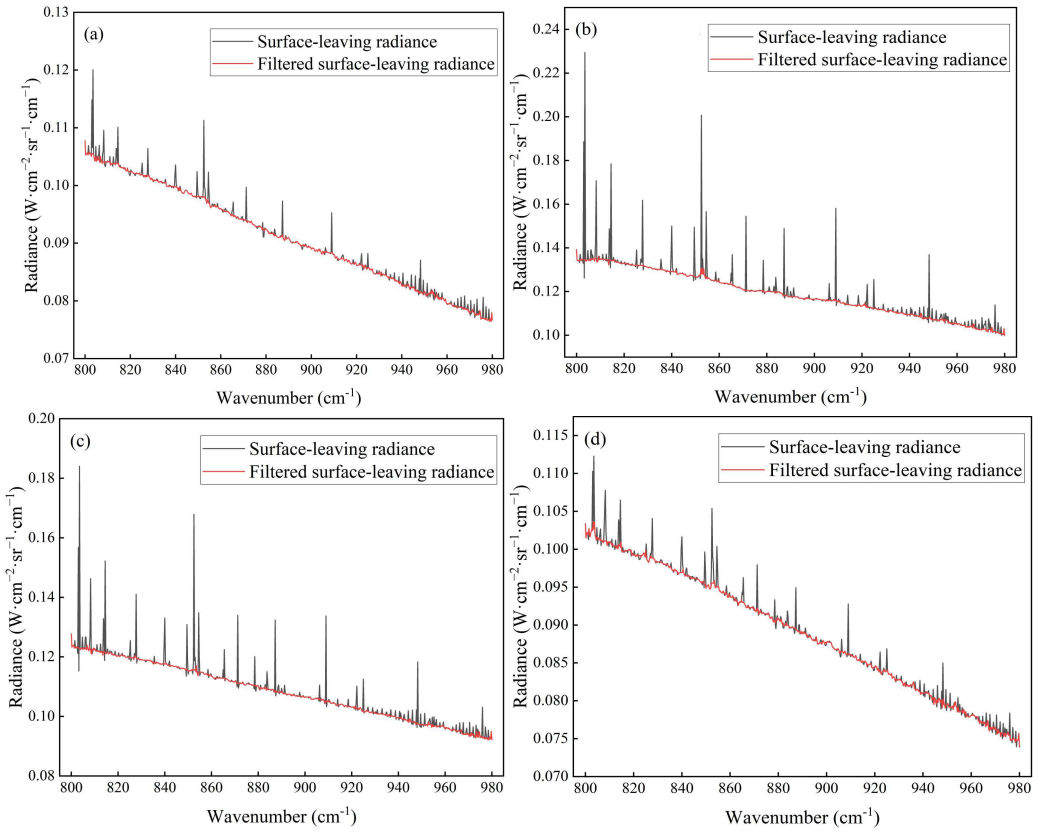


Fig. 16. Examples of comparison between surface-leaving radiances without and with Hampel filtering at four study areas. (a) Spain. (b) North Africa. (c) Hulunbuir. (d) Yellow Sea.

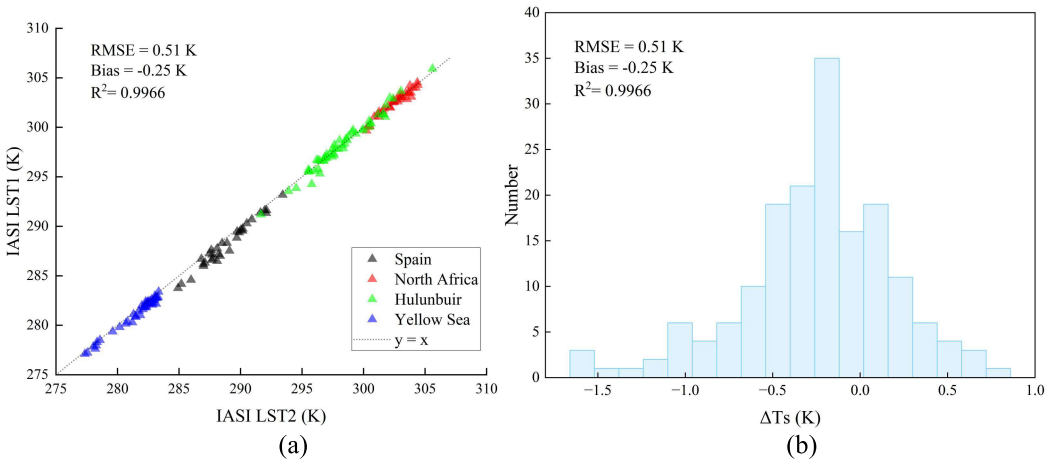


Fig. 17. Intercomparison results between retrieved LST (IASI LST1) and the published IASI LST (IASI LST2). (a) Scatterplot. (b) Histogram.

IASI LST1 and IASI LST2 is presented in Fig. 17(a), and the associated histogram of their differences, named ΔT_s, is presented as Fig. 17(b). It is worth noting that the RMSE is 0.51 K, with a bias of −0.25 K, while IASI LST1 ranging from 277 to 306 K; ΔT_s is predominantly distributed below 0.6 K, with only a limited number of outliers exceeding −1.5 K, with 80% concentrated between −1 and 0.5 K, indicating a general underestimation of LST. These results demonstrate higher accuracy than those reported by Zhang et al. [12], who retrieved LST completely by physical methods, and those by Chen et al. (2016) and Lan et al. [24], [25], who retrieved LST by directly constructing the relationship between the observed TOA BT and LST.

In addition, region-specific statistics of the LST retrieval results and histograms of ΔT_s are presented in Fig. 18 and Table VI, respectively. According to Table VI, North Africa exhibits the highest retrieval accuracy among the four regions, with an RMSE of 0.28 K and a bias of −0.06 K, likely due to the relatively homogeneous desert surface types in the region. In contrast, Spain shows the poorest performance, with an RMSE of 0.81 K and a bias of −0.68 K, which may be attributed to strong anthropogenic influences and high surface

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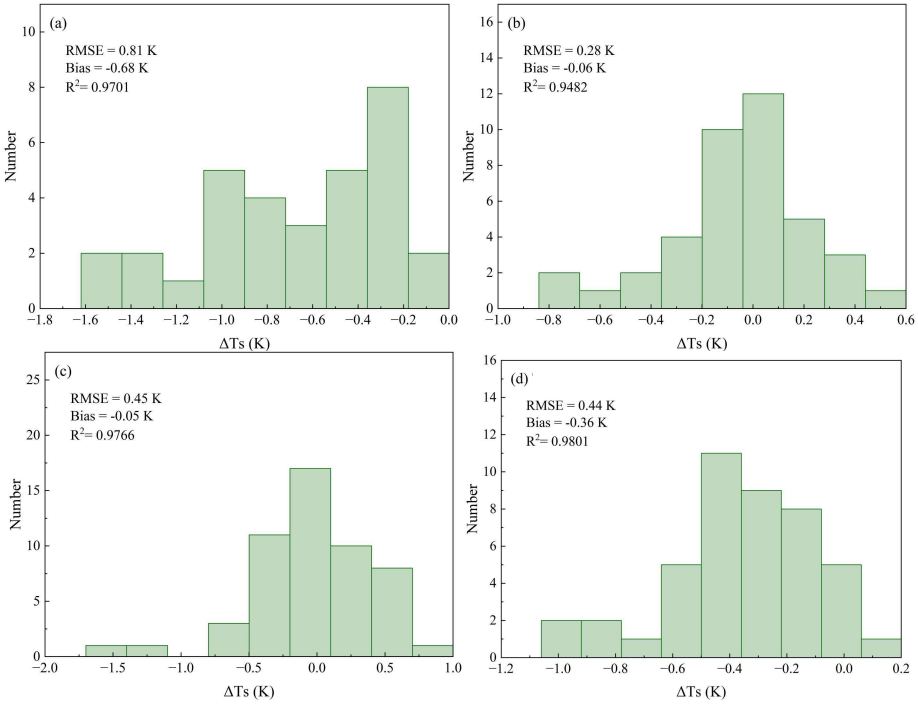


Fig. 18. Histograms of the difference between retrieved LST (IASI LST1) and the published IASI LST (IASI LST2) for four study areas. (a) Spain. (b) North Africa. (c) Hulunbuir. (d) Yellow Sea.

heterogeneity. As shown in Fig. 18, most retrieval results in Spain and the Yellow Sea exhibit underestimation, particularly for LST values ranging from 277 to 294 K.

In summary, the proposed LST retrieval algorithm effectively leverages the potential of hyperspectral TIR data by utilizing its abundant bands and spectral smoothing capabilities to mitigate the effects of residual atmospheric features on LST retrieval. When applied to IASI data, the algorithm produces retrieval results that closely align with the IASI L2 surface temperature products, demonstrating its robustness and practical applicability.

V. Conclusion

LST plays an essential role in understanding numerous climatic, ecological, and hydrological processes, making it a key parameter in fields ranging from climate science to agriculture. Hyperspectral TIR remote sensing offers distinct advantages over traditional multispectral sensors by providing high spectral resolution, which yields abundant information for retrieving both surface and atmospheric parameters. This study proposes a novel LST retrieval framework that combines a DR2N with the refined ISSTES algorithm improved by Hampel filtering. This method could effectively extract LST from IASI observed TIR data without the need for accurate atmospheric profiles. In this approach, the DR2N model is trained to perform atmospheric correction on simulated data generated by the 4A/OP radiative transfer model with TIGR atmospheric profile inputs, and validation on the simulated test dataset shows RMSEs of 0.115 K, 0.142 K, and 7×10^{-4} for upwelling radiance, downwelling radiance, and transmissivity, respectively. Subsequently, the surface-leaving radiance is smoothed by the Hampel filter for removing outliers, and the ISSTES algorithm is used to derive the LST. The result from simulated data over bare land, sea water, and grassland shows that the corresponding RMSE and bias between the retrieved and reference LSTs are 1.89 and −0.19 K, respectively, confirming the proposed algorithm's effectiveness.

To further validate its applicability, this method is applied to the IASI TIR observation over four diverse study areas, namely, Spain, North Africa, Hulunbuir, and the Yellow Sea. It is noted that in the spectral interval of 645–1600 cm^{-1}, the RMSEs for upwelling radiance, downwelling radiance, and transmissivity across all channels are 5.29 K, 3.00 K, and 0.016, respectively, while, in the 800–980-cm^{-1} range, the RMSEs are 4.55 K, 5.18 K, and 0.020, respectively. Following atmospheric correction, the LST is retrieved from IASI observed TIR data and is intercompared with publicly available IASI L2 surface temperature products, showing an overall RMSE of 0.51 K and a bias of −0.25 K. In general, the algorithm proposed in this study achieves results closely aligned with IASI L2 surface temperature products across diverse surface and atmospheric conditions. Meanwhile, results demonstrate high accuracy in LST retrieval for the three regions: North Africa (bare land), Hulunbuir (grassland and cultivated/managed land), and the Yellow Sea (sea water), achieving RMSE better than 0.50 K. The study area in Spain, which comprises cultivated and managed land, shows an RMSE of 0.81 K and a bias of −0.68 K, probably due to relatively high surface heterogeneity.

This study utilizes an extensive collection of satellite and simulated observations, encompassing a broad range of atmospheric conditions, surface types, and temperatures, to maximize the model's generalizability. However, real-world atmospheric and surface conditions are inherently more complex and dynamic. To further enhance the generalizability of the proposed deep learning model, future work should consider

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incorporating additional observation angles, expanding the simulated dataset to cover a wider range of surface and atmospheric scenarios, or integrating real observational data into the training process. Such improvements, however, would impose higher computational demands. For comparison, only IASI L2 products are used to minimize the potential impact of spatial–temporal inconsistencies. In the future, additional in situ measurements and those from MODIS and AIRS products over a wider variety of surface types could be incorporated for validation to further analyze the LST retrieval performance on different surface types. Furthermore, to permit more accurate assessment of the performances exhibited by the proposed retrieval methodology, conducting an uncertainty analysis of the comparison results represents a promising direction for future research. In particular, quantifying uncertainty within the DR2N model remains a challenging yet essential task for further development.

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